

YULIN LUO

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👤 RESEARCH PROFILE

I am a Ph.D. candidate in the School of Computer Science at Peking University, advised by Prof. Shanghang Zhang. My research focuses on open-world generalization for embodied intelligence, with emphasis on generalizable embodied foundation models, robot-scenario data and benchmark construction, and model-data co-evolution.

🎓 EDUCATION

Peking University

Sep. 2023 – Present

Ph.D. Candidate in Computer Science, School of Computer Science.

Advisor: Prof. Shanghang Zhang.

Shanghai Jiao Tong University

Sep. 2019 – Jun. 2023

B.Eng. in Automation, Department of Automation.

GPA: 3.87/4.30; Major rank: 2/89.

🔬 RESEARCH INTERESTS

- **Model:** building generalizable embodied foundation models for open-world robot intelligence.
- **Data/Benchmark:** constructing embodied-agent benchmarks, multimodal evaluations, and robot-scenario data.
- **Model-Data Co-evolution:** scaling embodied intelligence through data-driven model improvement and model-guided data construction.

📄 SELECTED PUBLICATIONS

1. Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models
Yulin Luo*, Hao Chen^{*,†}, Zhuangzhe Wu*, Bowen Sui*, Jiaming Liu^{*,†}, et al. *ACM MM'26 (CCF-A)*
2. RoboBench: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models as Embodied Brain
Yulin Luo*, Chun-Kai Fan*, Menghang Dong*, Jiayu Shi*, Xiangju Mi*, Mengdi Zhao^{*,†},
Bo-Wen Zhang*, Cheng Chi^{*,†}, et al. *ECCV'26 (CCF-B)*
3. LLM as Dataset Analyst: Subpopulation Structure Discovery with Large Language Model
Yulin Luo*, Ruichuan An*, Bocheng Zou, Yiming Tang, Jiaming Liu, and Shanghang Zhang. *ECCV'24 (CCF-B)*
4. Decomposing the Neurons: Activation Sparsity via Mixture of Experts for Continual Test-Time Adaptation
Rongyu Zhang*, Aosong Cheng*, **Yulin Luo***, Gaole Dai, Huanrui Yang, et al. *AAAI'26 Oral (CCF-A)*
5. RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization
Yiqing Shen*, **Yulin Luo***, Dinggang Shen, and Jing Ke. *MICCAI'22 (CCF-B)*
6. Efficient Deweather Mixture-of-Experts with Uncertainty-Aware Feature-Wise Linear Modulation
Rongyu Zhang, **Yulin Luo**, Jiaming Liu, Huanrui Yang, et al. *AAAI'24 (CCF-A)*
7. RoboMIND: Benchmark on Multi-Embodiment Intelligence Normative Data for Robot Manipulation
Kun Wu, Chengkai Hou, Jiaming Liu, et al., **Yulin Luo**, et al. *RSS'25 (CCF-B)*

* Equal contribution. † Project lead.

👜 RESEARCH INTERNSHIPS

Simplexity Robotics, Research Intern

Dec. 2025 – Present

- Research on embodied intelligence and generalizable robot foundation models.

- Beijing Academy of Artificial Intelligence (BAAI)**, Research Intern Aug. 2024 – Oct. 2025
- Research on embodied-brain benchmarks for multimodal large language models and robot intelligence.
- ByteDance**, Research Intern Mar. 2024 – Aug. 2024
- Trust & Safety Applied Algorithm Team for Ads and E-commerce.

HONORS AND AWARDS

- Shanghai Jiao Tong University Outstanding Graduate Jun. 2023
- Weichai Power Scholarship Oct. 2022
- Huawei Scholarship Dec. 2021
- Shanghai Jiao Tong University B-Class Merit Scholarship 2020, 2021, 2022

ACADEMIC SERVICE

- Reviewer for AAAI 2026, ICLR 2026, CVPR 2026, ECCV 2026, and NeurIPS 2026 Datasets & Benchmarks.